

Philippines

**Expanded National Nutrition Survey:
2018, 2019, & 2021**

BIOCHEMICAL COMPONENT

Data Documentation

December 2023

Expanded National Nutrition Survey: Biochemical Component

ENNS Management Team

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Overview

Producer	Department of Science and Technology – Food and Nutrition Research Institute (DOST – FNRI)
Identification	DDI-PHL-FNRI-UPS-2018-2019-2021-BSC-v01
Version	Production Date: 2023-12
Biochemical survey provides information on direct assessment of the nutritional status of a population as it determines levels of biomarkers such as hemoglobin (Hb), serum retinol (SR), and urinary iodine concentration (UIC) in blood and urine samples. The assessment is important in the early detection of nutrient deficiencies in the body before the appearance of clinical signs. Results obtained from biochemical assessment are utilized in conjunction with other nutritional assessment parameters (anthropometric, clinical, and dietary) to determine the overall nutritional status of an individual or population. The Biochemical Survey Component of the Expanded National Nutrition Survey (ENNS) provides data on biochemical indicators of Anemia, Iodine Deficiency, and Vitamin A Deficiency (VAD). Kind of Data Sample survey data	
Unit of Analysis	Individuals

Scope & Coverage

Country	Philippines
Geographic Coverage	National, 106 Provinces/ Highly Urbanized Cities (HUCs) for Iodine Deficiency while 96 Provinces/ Highly Urbanized Cities (HUCs) for Anemia and Vitamin A Deficiency

Population	Anemia Blood samples were collected from all population groups such as preschool children (6 months to 5 years), school-age children (6-12 years), adolescents (13-19 years), adults (20-59 years), older persons (60 years and over), and women of reproductive age (15-49 years) including non-pregnant/non-lactating, pregnant and lactating women. Vitamin A Deficiency Blood samples were collected from selected population groups such as preschool children (6 months to 5 years), older persons (60 years and over), and women of reproductive age (15-49 years) including non-pregnant/non-lactating, pregnant and lactating women. In 2021, school-age children (6-12 years) were included in the collection of blood samples for Vitamin A Deficiency. Iodine Deficiency Urine samples were collected from selected population groups such as children (6-12 years), older persons (60 years and over), and women of reproductive age (15-49 years) including non-pregnant/non-lactating, pregnant and lactating women.
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Sampling	
Sampling Procedure	The National Nutrition Survey (NNS) survey design was changed to become a rolling survey design and was termed as Expanded National Nutrition Survey (ENNS) in 2018. The ENNS had a wider scope of coverage and had a longer time frame of data collection to cover all 81 provinces, 33 HUCs (including 16 cities in the National Capital Region), and 3 other urban areas (Pateros, Isabela City, and Cotabato City), from 2018 to 2020. The Expanded National Nutrition Survey (ENNS) was conducted in 2018, 2019, and 2021. It utilized a rolling sample design in which 117 Philippine Statistics Authority (PSA) domains of provinces and highly urbanized cities (HUCs) were grouped into 24 replicates having similar characteristics utilizing data from the 2010 Census of Population and Housing (CPH). Eight replicates consisting of 40 domains were independently allocated each year to generate national-level estimates. For the regional estimates, the total provinces/HUCs covered in the three periods of the ENNS were cumulated to come up with the regional level data of the different nutrition and health indicators collected in the survey. In each domain, the selection of households utilized the fully replicated design of the 2013 PSA Master Sample (MS) design. All 16 replicates consisting of MS enumeration areas (EAs) were fully covered which allows for more reliable estimates at the domain level. The number of EAs comprising each MS replicate depends on the size of the domain. In 2018, 40 provinces and HUCs were surveyed; while in 2019, only 39 provinces, HUCs, and other urban areas were covered since the Province of Sulu was not surveyed due to the peace and security concerns in the area. The 2018 and 2019 ENNS results were pooled to increase the precision for tracking trends of nutrition and health indicators. Lastly, for the third year of the ENNS that was interrupted in 2020 due to COVID-19 pandemic, 37 areas were covered from July 2021 to June 2022. Only 50% of the target households were covered for the biochemical component due to the high cost of laboratory analyses.

Sample Size	Hemoglobin	149,045
	Serum Retinol	68,530
	Urinary Iodine Concentration	90,519

Methodology	
Data Collection	Blood Samples Blood samples were collected by trained registered medical technologists from preschool children, 6 to 60 months old via the finger prick method using sterile blood lancets while the venipuncture method was used for study participants 61 months and over using sterile syringes and needles. Hemoglobin, a biomarker of anemia was analyzed in the field using the cyanmethemoglobin method (ICSH, 1978). Serum retinol, a biomarker of Vitamin A status, was measured using the High-Performance Liquid Chromatography (HPLC) and this was done at the DOST-FNRI Laboratory. Urine Samples About 5 ml of midstream urine sample was collected in a clean plastic cup among children (6-12 years), older persons (60 years and over), and women of reproductive age (15-49 years) including non-pregnant/non-lactating, pregnant, and lactating women. Urine samples were collected to test for the Urinary Iodine Concentration (UIC) to provide data on the iodine status. UIC was determined at the DOST-FNRI Laboratory using the ammonium persulfate/uv-vis method.
Data Entry System	Initial editing and validation of data were performed directly in the e-DCS by the field researchers. A backup copy of the data was also submitted to the Team Leader via USB flash drive in which another round of data validation was performed. Upon online transmittal to the central database, final data editing and validation were done for all interview-based survey questions.
Data Processing and Statistical Analysis	At the DOST-FNRI, the encoded data was checked against the edited survey questionnaires. Blood pressure measurements were checked by cross-referencing the printed copy of the encoded measurements to the handwritten values in the hard copy/printed forms. Several rounds of alternating manual and machine validations were done until the desired consistency and completeness of data were achieved. Errors or outliers found were validated with the actual entries/answers in the survey questionnaires. Errors in responses/answers were excluded. Sampling weights were computed based on the product of the base weights, non-response adjustment, and post-stratified calibration/adjustment based on the projected population counts obtained from the PSA. The application of the National Sampling Weights can only be used to estimate national-level data of each ENNS survey year (2018, 2019, and 2021). Alternatively, other than estimating provincial or domain-level data, Provincial Sampling Weights can also be used for the national-level data estimation given that the ENNS survey years are combined. However, the Provincial Sampling Weights for Biochemical Component can only be used for the estimation of provincial or domain-level data and cannot be used for the national-level estimation of combined three survey years due to the partial coverage of domains in the 2021 ENNS. Stata versions 15 and 16 were used to generate descriptive statistics including

	mean, median and percentage, range, confidence intervals, design effect, and coefficient of variation. Since the third and last year of the ENNS was disrupted in 2020, data from the 2018 and 2019 ENNS were pooled to increase the precision of national-level estimates and to further improve comparability vis-a-vis results from previous NNS surveys.
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Notes

The researchers who collected the survey data were grouped into sub-teams. In 2018 and 2019 ENNS, each sub-team was composed of two team leaders and eight members. However, the 2021 ENNS sub-teams only consist of one team leader and five members. These sub-teams stayed in the survey area for at least one week to perform the tasks, including the location of the sample households, data collection, editing, and data transmission and transfer to the next area.

For the 2021 ENNS, all the mandatory health and safety protocols as directed by the Department of Health and Local Government Units (LGUs), such as wearing face masks and face shields, conducting physical distancing, and frequent hand washing or sanitizing of hands, were strictly implemented, to protect the health of both the researcher and the respondent against the COVID-19.

Variable Label and Definition		
Biochemical Component		
Variable Name	Variable Label	Value Labels/Variable Definition
regcode	Name of Regions	Refers to the different regions in the country which were coded accordingly:
	Codes given to Regions	1 Ilocos
		2 Cagayan Valley
		3 Central Luzon
		5 Bicol
		6 Western Visayas
		7 Central Visayas
		8 Eastern Visayas
		9 Zamboanga Peninsula
		10 Northern Mindanao
		11 Davao Region
		12 SOCCSKSARGEN
		13 National Capital Region
		14 Cordillera Administrative Region
		15 ARMM
		16 Caraga
		41 CALABARZON
		42 MIMAROPA
provhuc	Province and HUCs Level	Refers to the different provinces and HUCs that were covered in the ENNS:
	[Sulu, was not covered due to the peace and security concerns]	1 Ilocos Norte 2 Ilocos Sur 4 Pangasinan 5 Batanes

Variable Label and Definition		
Biochemical Component		
Variable Name	Variable Label	Value Labels/Variable Definition
	in the area]	6 Cagayan Province 7 Isabela Province 8 Nueva Vizcaya 9 Quirino 10 Bataan 11 Bulacan 12 Nueva Ecija 13 Angeles City 14 Pampanga 15 Tarlac 16 Zambales 17 Olongapo City 18 Aurora 21 Laguna 22 Quezon Province 23 Lucena City 24 Rizal 25 Marinduque 27 Oriental Mindoro 28 Palawan 29 Puerto Princesa City 30 Romblon 31 Albay 32 Camarines Norte 33 Camarines Sur 34 Catanduanes 35 Masbate 36 Sorsogon 37 Aklan 38 Antique 39 Capiz 40 Iloilo City 41 Iloilo Province 44 Guimaras 45 Bohol 46 Mandaue City 47 Lapu-Lapu City (Opon) 48 Cebu 49 Cebu City 50 Negros Oriental 51 Siquijor 52 Eastern Samar 53 Tacloban City 54 Leyte 55 Northern Samar 56 Western Samar 57 Southern Leyte 58 Biliran 59 Zamboanga del Norte 60 Zamboanga City 61 Zamboanga Del Sur

Variable Label and Definition		
Biochemical Component		
Variable Name	Variable Label	Value Labels/Variable Definition
		62 Zamboanga Sibugay
		63 City of Isabela
		64 Bukidnon
		65 Camiguin
		66 Lanao Del Norte
		67 Iligan City
		68 Misamis Occidental
		69 Cagayan De Oro
		70 Misamis Oriental
		71 Davao Del Norte
		72 Davao City
		74 Davao Oriental
		75 Compostela Valley
		76 Davao Occidental
		77 North Cotabato
		78 South Cotabato
		79 General Santos City
		80 Sultan Kudarat
		81 Sarangani
		82 Cotabato City
		83 City of Manila
		84 Mandaluyong City
		85 San Juan City
		86 Quezon City
		87 City of Marikina
		88 City of Pasig
		89 Caloocan City
		90 City of Malabon
		91 City of Navotas
		92 City of Valenzuela
		93 Makati City
		94 Taguig City
		95 Las Piñas City
		96 City of Parañaque
		97 Pateros
		98 City of Muntinlupa
		99 Pasay City
		100 Abra
		101 Baguio
		102 Benguet
		104 Kalinga
		105 Mountain Province
		107 Basilan
		108 Lanao Del Sur
		109 Maguindanao
		111 Butuan City
		112 Agusan Del Norte
		113 Agusan Del Sur
		114 Surigao Del Norte
		115 Surigao Del Sur
		116 Dinagat Islands

Variable Label and Definition Biochemical Component		
Variable Name	Variable Label	Value Labels/Variable Definition
enns_year	ENNS Year	ENNS Years 2018 ENNS conducted from 2018 to 2019 2019 ENNS conducted from 2019 to 2020 2021 ENNS conducted from 2021 to 2022
hhnum	Household Number (Unique household identifier; merging variable)	A unique number code is assigned to each of the sample households. Each individual belonging to the same household has a similar household identifier. This variable was used to merge variables from different nutrition survey components.
member_code	Member Code (Unique identifier; merging variable)	It is a pre-coded membership number of household members.
fwgti_natl2_var	Sampling Weights National	Product of the base weights, non-response adjustment, and post-stratification adjustment for yearly national-level estimation (2018, 2019, and 2021)
fwgti_prov2	Sampling Weights Provincial	Product of the base weights, non-response adjustment, and post-stratification adjustment for provincial-level estimation
rep_natl	ENNS Replicate Codes	Replicate codes assigned to PSA domains for national-level estimation (1,2,...,24)
rep_prov	MS Replicate Codes	Replicate codes assigned to PSA MS EAs/PSU for domain/provincial-level estimation (1,2,...,16)
ms_psucode	MS PSU Codes	PSU code based on the PSA Master Sample (MS)
uic	Level of UIC (micrograms per liter)	Urinary iodine concentration (UIC) was used to determine the level of iodine intake in a population and to classify adequacy or inadequacy of intake based on WHO/UNICEF/ICCIDD cut-offs.
vita	Level of Vitamin A (micrograms per deciliter)	Serum retinol was used to measure the concentration of Vitamin A and to determine the prevalence of VAD based on WHO cut-offs.
hemoglobin	Level of Hemoglobin (grams per deciliter)	Hemoglobin was used to determine the prevalence and to classify the public health significance of anemia based on WHO cut-offs.

Accessibility	
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